



Every email accounted for. Every shipment answered for.

A **bill of lading** is the ship's receipt for a container – what is inside, whose it is, where it is going. We check every draft against what the customer actually asked for, and catch the shipments nobody ever emailed about.

- Two independent checks, not one sorting model
- Every verdict clicks back to its source line
- Refuses rather than guesses

TEAM T010NG

@kymil4

Backend · Pipeline · Frontend

@AlaskanTuna

Fullstack · DevOps · Cloud

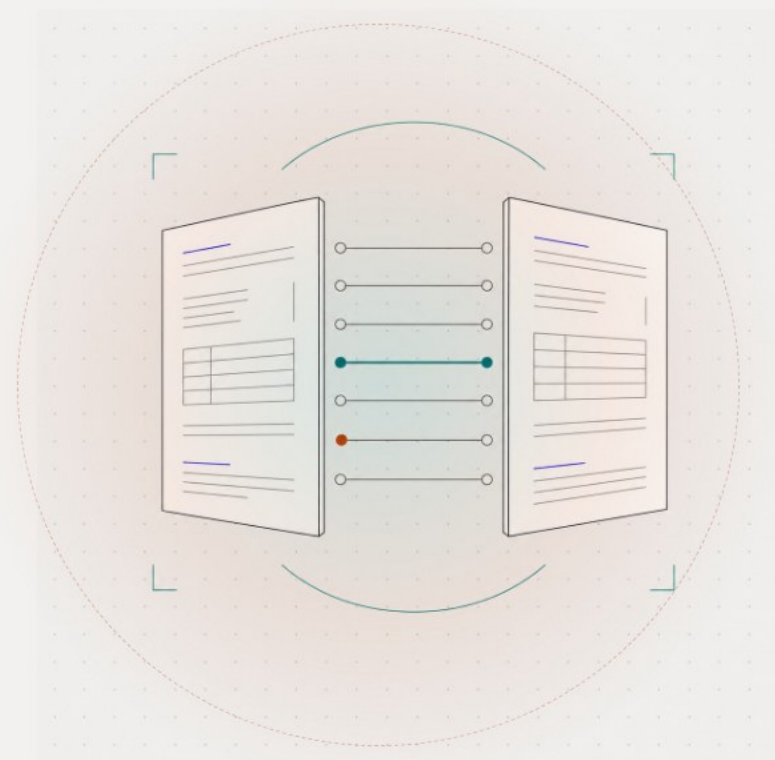
@chaosiris

Persistence · Deployment ·
Verification

@DrxgClanPC

Ideation · Pitch · Review

ALIGNED WITH UN
GLOBAL GOALS 8
AND 9



averis-222536409832.asia-southeast1.run.app/judge
github.com/Averis-T010NG/LadingLens
Synthetic data only · no account required

One request. Seven fields. Two of them are wrong.

Two thousand emails can land here in a day. This is one of them.

EMAIL 065 • FROM THE ORGANISERS' BUNDLE

**One 40-foot container of paper.
Port Klang → Ho Chi Minh City.**

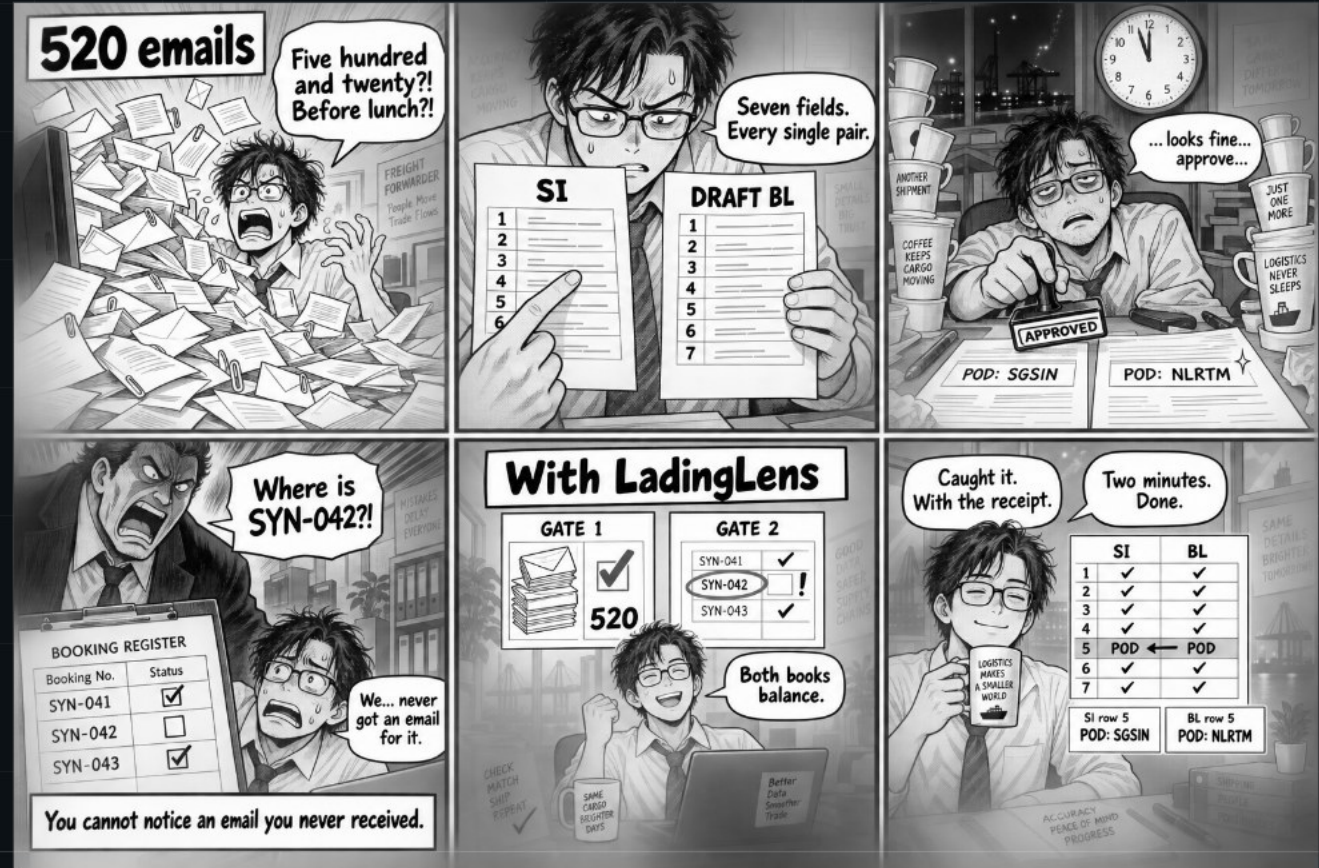
Where the goods are unloaded

SI Ho Chi Minh City, Vietnam (VNSGN)
BL **Busan, South Korea** (VNSGN)

Who gets told on arrival

SI International Forest Products LLC
BL **Habras International Limited**

Ports have short codes, like airports. VNSGN is Ho Chi Minh – printed beside the word Busan. Either the city is wrong or the code is, and nothing here says which. So it is flagged for a person, not signed.



WHAT ONE MISS CAN COST

MYR 200 amendment fee

Up to MYR 5,000 customs fine

Demurrage, or a rolled sailing

Volume from the Averis brief • fee and fine from Maersk Malaysia guidance, for an amendment inside five days of arrival

Sort the inbox perfectly and **one failure still walks straight through.**

Classification is a complete answer to the wrong question. It can only ever account for what arrived.

WHAT A CLASSIFIER CAN SEE

Every message that arrived

Sorted, labelled, routed and counted

Every attachment it carried

Read, extracted and compared field by field

Every case it opened

Tracked through to a verdict and an owner

Every field it compared

Seven per pair, each one anchored back to a line

A perfect score on all three, and a container can still sail on paperwork that nobody ever raised.

WHAT IT CANNOT

You cannot notice an email you never received.

No message

Nothing to misfile

No attachment

Nothing to misread

No case

Nothing at all to flag

A booking with no email

An amendment never sent

A document that never left the shipper

The only way to see an absence is to check the inbox against a **second, independent record** of what should have arrived.

THEREFORE

One record of what arrived

One record of what was expected

Reconcile the two, every run

The gap between those two records is the entire product

Borrow the answer from **double-entry** bookkeeping.

Two independent books. Checked separately, then reconciled against each other — so neither can quietly cover for the other.

GATE 1 • WHAT ARRIVED

Every email is accounted for

Receipted and hashed before anything else runs, then given exactly one of five categories by pinned jev-1.13.0.

THE ARITHMETIC THAT HAS TO HOLD

520 received

520 accounted for

0 lost

Nothing is silently dropped. Every message ends on a recorded outcome, including the ones a person has to decide.

GATE 2 • WHAT SHOULD HAVE ARRIVED

The expected-shipment ledger

MISSING_CASE

On the ledger, no case exists

UNMATCHED

A case with no ledger entry

AMBIGUOUS

Both candidate sets kept

An independent ledger reconciled against the cases that exist — never derived from the inbox, which is the entire point.

SYN-042 expects a draft BL and has no case. Gate 1 could never raise it; there was no email to raise.

WHY TWO

An inbox proves what came

A ledger proves what should have

Only the pair proves absence

Two independent controls, one reconciliation

Every verdict points back at the **exact line** it came from.

Seven fields, with the Shipping Instruction always the reference. No value enters a verdict without an anchor into its source.

THE RULE

No anchor, no answer.

A provenance anchor is not a citation added afterwards. It is the condition on which a value is allowed to exist at all.

Reference

The Shipping Instruction, always

Compared

Against the draft Bill of Lading

A value that cannot be grounded **fails the run** rather than entering a verdict.

WHAT AN ANCHOR IS, BY FORMAT

TXT

Line number, plus Unicode code-point start and end columns

XLSX

Sheet name, plus an exact A1 cell reference

DOCX

Table cell coordinates, or the paragraph index

PDF

Page, plus a 72-dpi bounding box when the PDF is digital

THE SEVEN COMPARED FIELDS

Shipper

Consignee

Notify party

Port of loading

Port of discharge

Container count

Gross weight

Chinese labels included — the columns are **code-point exact**, not byte offsets, so a highlight lands on the character a reader can actually see.

ONE CLICK

Open a verdict

Land on the line

Download the artifact

Seven fields • every one of them anchored, or refused

The question is not what AI can read. It is **who is allowed to decide.**

Four decisions, four owners, stated up front — and two rules that hold when a provider lets us down.

THE REGISTER

Reads documents

Gemini 3.5 Flash, grounded to anchors — scanned and ambiguous documents only

Judges meaning

Pinned jev-1.13.0 — category, document role, field equivalence, typed answers only

Computes

Deterministic code — parsing, normalisation, numbers, schema, state transitions

Decides

A named human — anything consequential or unsafe to automate

RULE ONE

Fail closed

A timeout, an exhausted quota, an invalid typed answer, or a value that cannot be grounded is recorded as a failure — never as a result.

RULE TWO

Never fabricate

No fallback provider, no guessed category. A case with no answer shows as needing review, with a retry.

Seventeen of 520 cases are refused rather than guessed. That count is the design working.

REGISTER

docs/ai.md carries every decision and its owner

Gemini 3.5 Flash

jev-1.13.0, pinned

Automatic where the evidence is sufficient, human-owned where it is not

A public path. **No account.** Live against the real providers.

Nothing here needs a credential, a fixture or a rehearsal — the link stays open for the whole judging window.

THE WORKSPACE AT /JUDGE

Synthetic data only. Do not upload real shipping documents.

Judge workspace

Shipping Instruction

Drop files here or press Enter to browse
Accepts TXT, PDF, DOCX, XLSX up to 5.24288 MB

Draft Bill of Lading

Drop files here or press Enter to browse
Accepts TXT, PDF, DOCX, XLSX up to 5.24288 MB

☐ These documents are synthetic (no real shipping data)

Demo dataset summary ⓘ

PREPARED BASELINE

520 of 520 emails accounted for

OK (457) Mismatch (46) Needs review (17)

Case present (1) Document missing (1) Missing case (1) Unmatched case (125) Duplicate or ambiguous (1) Source stale (1)

Demo artifacts

[Open the inbox](#) [Open reconciliation](#) [Open the example case](#)

FIVE STEPS, ABOUT FOUR MINUTES

01 Open the link

A guest session mints itself — no sign-in, no email

02 Drop a pair

Any shipping instruction and draft BL that you bring

03 Watch it run

Real providers, in real time — never a fixture

04 Read the verdicts

Click any of the seven through to its source line

05 Take the artifacts

Download the run, hit Reset All, and then do the whole thing over again

On provider failure the upload stays retryable, and any prepared example appears only under a **PREPARED FALLBACK** label.

OPEN PATH

No account

No allow-list

Synthetic data only

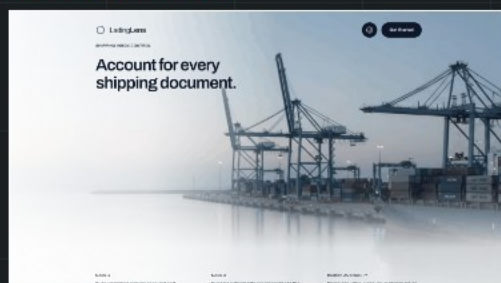
Reset between runs

Reachable without sign-in throughout the judging window

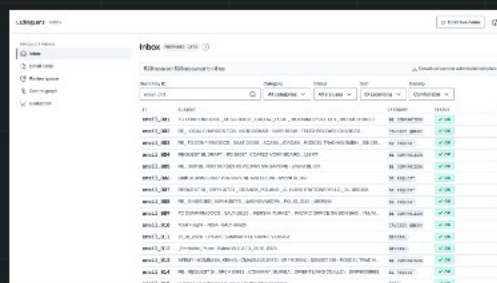
Five minutes, **end to end**, against the deployed service.

Every frame below is the running deployment, captured at 1440×900. None of it is a mock-up.

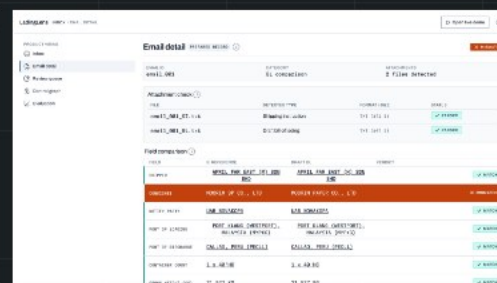
THE WALKTHROUGH



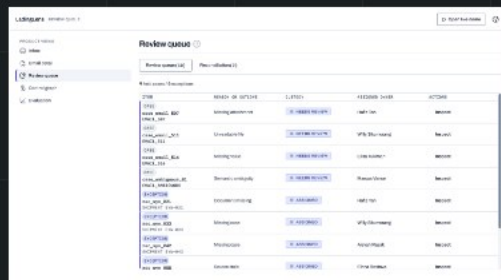
01 GUEST ENTRY, NO ACCOUNT



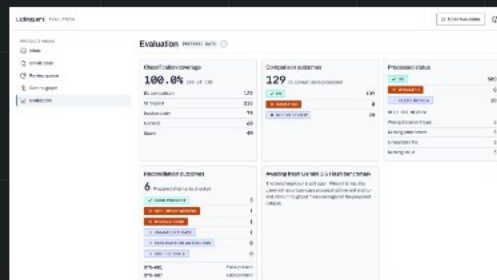
02 GATE 1 • 520 ACCOUNTED FOR



03 SEVEN FIELDS, CLICK TO SOURCE



04 GATE 2 • SYN-042 MISSING



05 THE RUN, SCORED END TO END



06 YOUR OWN PAIR AT /JUDGE

FULL DEMO VIDEO



Scan to run it
yourself, live

The recorded walkthrough goes to the submission
form before the deadline.

Tracked in **issue #46**

Not yet recorded at time of writing

CAPTURED ON

The deployed Cloud Run service

Synthetic bundle

No staged screens

Six frames, in the order a judge would meet them

Every message carries an outcome **on the record**.

Five hundred and twenty messages in, five hundred and twenty accounted for. This is where each one of them landed.

ALL 520, BY OUTCOME

457

OK
87.9%

46

MISMATCH
8.8%

17

NEEDS REVIEW
3.3%

520

ACCOUNTED FOR
0 LOST

Nothing dropped

Every inbound message ends in a state a person can name out loud

Nothing invented

No gap is filled with a plausible guess to keep a number tidy

The 46

A field differs between the two documents – flagged with both values and both anchors

The 17

The evidence was not sufficient to decide, so nobody pretended that it was

Four hundred and fifty-seven clean, forty-six flagged, seventeen refused. That refusal count is the design working, not a shortfall in it.

HOW THE COUNT IS REACHED

Gate 1 · Completeness

520 received, 520 accounted for, 0 lost

Gate 2 · Reconciliation

SYN-042 expects a draft BL and has no case at all – MISSING_CASE, structurally invisible to the inbox

Held for a person

Unreadable draft BL, blank customer fields, wrong document type – handed on by name, never guessed

By design

The refusal rate is published beside the pass rate, never folded into it

Gate 2 is the only check that can find **SYN-042**. No amount of accuracy on Gate 1 would ever have surfaced it.

BASELINE

Organisers' synthetic bundle

Prepared seed, labelled as prepared

Re-runnable from the repository

Prepared seed baseline from the organisers' synthetic bundle

One Cloud Run service. No long-lived credentials.

FastAPI and the compiled React build ship as one origin, deployed keylessly, and every deploy has to pass a fail-closed smoke check before it counts.

RUNTIME

One origin, no CORS surface

Python 3.12

FastAPI, SQLAlchemy 2 async

React 19

TypeScript 6, Vite 8, Bun

PostgreSQL 16

Append-only audit via DB triggers

Private GCS

Evidence bytes, never public

The demo link is a single origin, so there is no cross-origin surface to get wrong.

DEPLOY & VERIFY

Keyless CI, fail-closed gate

GitHub Actions authenticates through Workload Identity Federation, pinned to this repository's numeric id and main. Secrets reach the runtime only through Secret Manager – no service-account key is stored anywhere.

EVERY DEPLOY MUST PASS

Health

Readiness on a real DB

SPA fallback

Unauthenticated /judge

520-record artifact

Private-object denial

If any check fails the deploy is marked failed and the JSON report is retained as a build artifact.

BOUNDARY

Evidence served only through authorized endpoints

Never a public bucket URL

The measured numbers – including the one we missed.

A demo that hides its worst number is worth less than one that names it. Everything below is checkable in the repository.

THE MISS

25.6s

LIVE-PATH P95
AGAINST A 10 S
TARGET

Three warm-ups then twenty measured trials on one scanned pair. Only 5 of 20 completed.

No latency claim is made beyond what the retained artifact shows.

Run 35579538701 · commit 4eb1401

SCOPE OF THE PRELIMINARY BUILD

Synthetic only

DATA_POLICY=synthetic-only, enforced server-side

Guest entry

Sign-in fields are presentational, never sent or stored

Prepared baseline

The inbox is seeded, and labelled as prepared

Live path

Only /judge uploads run the real providers

A provider failure fails closed with a plain message and a labelled **PREPARED FALLBACK** – never a fabricated result.

SOURCES

[docs/ai.md](#)

[Retained benchmark artifact](#)

[Nothing cited that is not measured](#)

Close the loop, then **earn the right** to real data.

Ordered by what unblocks the next honest claim, not by what demonstrates best.

NEAR • IN FLIGHT NOW

Make the live path fast enough

Concurrent per-document calls and a warmer extraction path, measured in exactly the same way, until p95 sits under the target rather than beside it.

Concurrency

Per-document calls issued together instead of in series

Warm start

The extraction path kept warm between runs

Same method

Three warm-ups, twenty measured trials, one artifact retained – so the next figure is comparable to this one

Until that number moves, the **25.6 s** on the previous slide is the only latency figure we will quote.

AFTER THAT, IN ORDER

Reconcile the real ledger

Swap the synthetic expected-shipment CSV for the operational source, so Gate 2 reports on live bookings instead of on a fixture

Earn real documents

Retention, access, transfer and provider controls approved before a single production document is processed

Operational feed

Gate 2 on live bookings

Signed off, not merely convenient

Each step unblocks the claim after it. None of the three is skippable.

NOT YET

Real customer documents

A published latency claim

Anything a person has not signed off

Each step unblocks the claim after it

Do not take our word for **any** of it.

The point of all of it is a freight desk that works by exception, not by volume. Two links – one runs the system against a pair we have never seen, the other is every line that makes it work.

WHAT TO DO WITH THEM

Bring your own pair

Upload a shipping instruction and draft BL we have never seen

Audit a verdict

Click any one of the seven through to the exact line it came from

Check our arithmetic

Download the artifact, reset the workspace, and run it again

Everything claimed in this deck is either on screen at one of those two links, or it is not claimed.

WITH REGARDS,

– *Team T010NG*

SCAN EITHER ONE



LIVE • /JUDGE

...run.app/judge



GITHUB

Averis-T010NG/LadingLens

Both links stay up for the whole judging window, and neither asks for an account.